

K/U	A	T	C	Total
15	16	17	12	60

Which of the following is NOT true for a system at equilibrium

Equilibrium can only be reached in an open system

the rate of the forward reaction equals the rate of the reverse reaction

equilibrium may be reached from a forward or reverse reaction

all species must be present in a constant concentration

Which of the reactions with the following K_c 's represent an equilibrium with the greatest concentration of reactants:

$$3.42 \times 10^3$$

$$342000$$

$$0.342$$

$$3.42 \times 10^{-5}$$

$$3.42 \times 10^{-1}$$

For a specific reaction, which of the following statements can be made about the equilibrium constant (K_c)?

It always remains the same under any set of conditions

It increases when the concentration of one of the products is increased

It changes with changes in the temperature

It increases when the concentration of one of the reactants is increased

It can be changed by the addition of a catalyst

Decreases in temperature, with volume and pressure constant, tend to:

shift an equilibrium system in favour of products for an exothermic process

shift an equilibrium system in favour of products for an endothermic process

shift an equilibrium system in favour of the side in which fewer gas molecules are present

have no effect on an equilibrium system

During an experiment, a student pours 25.0 mL of 1.40 mol/L nitric acid, $\text{HNO}_{3(aq)}$, into a beaker that contains 15.0 mL of 2.00 mol/L sodium hydroxide solution $\text{NaOH}_{(aq)}$. The resulting solution would be...

- a) Basic
- b) Acidic
- c) Neutral
- d) Impossible to determine without the K_a of HNO_3
- e) Impossible to determine without the K_b of NaOH

Which statement for the ion product constant for water is valid?

a)	$K_w = K_a K_b$	d)	$K_w = [\text{H}_3\text{O}^+][\text{OH}^-]$
b)	$14 = \text{pH} + \text{pOH}$	e)	all of the above are correct
c)	$K_w = 1.0 \times 10^{-14}$		

What is the conjugate acid of $\text{H}_2\text{PO}_4^-_{(aq)}$?

a)	$\text{H}_3\text{O}^+_{(aq)}$	d)	$\text{PO}_4^{3-}_{(aq)}$
b)	$\text{OH}^-_{(aq)}$	e)	$\text{H}_2\text{PO}_4^-_{(aq)}$
c)	$\text{H}_3\text{PO}_{4(aq)}$		

A solution contains 1.78×10^{-11} mol/L of hydroxide ions $[\text{OH}^-]$. What is the pH of the solution?

a)	3.25	d)	10.75
b)	6.51	e)	none of the above
c)	7.49		

Which of the following would not undergo 100% dissociation (equilibrium completely to the right) in water

a)	NaOH	d)	H ₂ SO ₄
b)	CH ₃ COOH	e)	HClO ₄
c)	KOH		

A solution contains 1.58×10^{-10} mol/L of hydronium ions [H₃O⁺]. What is the pH of the solution?

- a) 4.20 d) 9.80
b) 5.00 e) 8.10
c) 6.30

What is the solubility product constant expression for calcium phosphate (Ca₃(PO₄)_{2(s)})?

- [Ca²⁺]²[PO₄³⁻]³
[Ca²⁺]³[PO₄³⁻]²
[Ca²⁺][PO₄³⁻]
[Ca²⁺][PO₄³⁻]/[Ca₃(PO₄)₂]
1/[Ca²⁺]²[PO₄³⁻]³

Which of the following acids would have the strongest conjugate base?

- a. HCN Ka = 6.2×10^{-10} CN⁻ Kb = ?
b. HCOOH Ka = 1.8×10^{-4} HCOO⁻ Kb = ?
c. HNO₂ Ka = 7.2×10^{-8} NO₂⁻ Kb = ?
d. HF Ka = 6.6×10^{-2} F⁻ Kb = ?

e. All acids are the same strength, as long as they have a dissociable hydrogen ion.

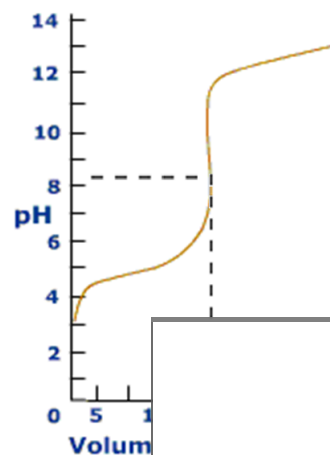
Which of the following conjugate acid-base pairs would not act as a suitable buffer system?

- a. CH₃COOH and CH₃COONa
b. NH₃ and NH₄Cl
c. HCl and NaCl
d. NaHCO₃ and Na₂CO₃

Looking at the graph on the right, indicate which of the following reactants were used to generate this graph

- a. A weak base titrated with a strong acid
b. A weak acid titrated with a strong base
c. A strong acid titrated with a strong base
d. A strong base titrated with a strong acid

Which of the following indicators would be most useful



for the titration graph on the right

pH Range (range where colour will change)

- | | | |
|--------------------|----------------------|-------------|
| a. methyl orange | [red to orange] | [4.4-3.1] |
| b. cresol red | [yellow to purple] | [12.0-10.1] |
| c. methyl red | [yellow to red] | [6.2-4.3] |
| d. phenolphthalein | [colourless to pink] | [10.0-8.2] |

Part A: Multiple Choice [15 marks]

1.			6.			11.				
2.			7.			12.				
3.			8.			13.				
4.			9.			14.				
5.			10.			15.				

60

Part B: Fill in the Blank [17 marks] (*spelling does not count as long as I know what you are trying to say*)

During an equilibrium process when all reactants and products are present in the same phase we call this a _____.

During an equilibrium process when the reactants and products are in different phases we call this a _____.

A buffer solution _____ changes in pH when a moderate amount of an acid or a base is added to it.

An acid is a substance that dissociates in water to produce one or more H^+ ions and a base is a substance that dissociates to form one or more OH^- ions. This is known as the _____ theory of acids and bases.

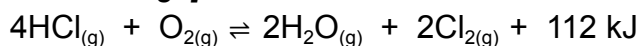
An acid is a substance from which a proton (H^+ ion) can be removed and a base is a substance that can accept a proton (H^+ ion) from an acid. This is known as the _____ theory of acids and bases.

A _____ acid will completely dissociate in water while a _____ acid will partially dissociate in water.

The following statement "A dynamic equilibrium tends to respond to relieve the effect of any change in the conditions that affect the equilibrium" is known as _____.

The _____ occurs in a titration when the indicator changes colour, which happens over a range of ~2 pH units while the _____ is the moment when all the acid and base that are present in a titration have completely reacted with one another.

What effect would the following changes have on the equilibrium concentration for $[\text{Cl}_2]$?
[increase, decrease or no change]



- a) Increasing the temperature of the reaction vessel _____
- b) Increasing the total pressure (decreasing volume) _____
- c) Increasing the $[\text{O}_2]$ _____
- d) Increasing the volume (decreasing pressure) _____
- e) Adding a catalyst _____
- f) Removing $[\text{H}_2\text{O}]$ _____
- g) Adding an inert gas $[\text{Ar}]$ _____

Part C: Short Answer [16 marks]

SHOW ALL YOUR WORK!!

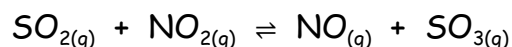
The solubility of iron (III) hydroxide ($\text{Fe}(\text{OH})_{3(\text{s})}$) is $3.40 \times 10^{-6} \text{ mol/L}$.

Calculate the solubility product constant (K_{sp}) for this compound. (3 marks)

Using your K_{sp} value from above, calculate what the new **molar solubility** would be for this compound if 0.10 mol/L NaOH was present in solution already.

(3 marks). If you do not have a value for part a, use a K_{sp} value between $(\# \times 10^{-20}$ to $\# \times 10^{-25})$

A chemist was studying the following reaction.



Part A: In a 1.0 L container, the chemist added 1.7×10^{-1} mol of $SO_{2(g)}$ to 1.1×10^{-1} mol of $NO_{2(g)}$. The value of K_c for the reaction at a certain temperature is 4.8. What is the equilibrium concentration of all reactants and products at this temperature? (6 marks)

Part B: For the reaction above, if the following concentrations are given, calculate the value for Q_c and determine if the reaction is at equilibrium, if not, state which direction the reaction will proceed to [4 marks]

a) $[\text{SO}_2] = 0.20 \text{ mol/L}$ $[\text{NO}_2] = 1.00 \text{ mol/L}$ $[\text{NO}] = 1.50 \text{ mol/L}$ $[\text{SO}_3] = 3.00 \text{ mol/L}$

b) $[\text{SO}_2] = 1.20 \text{ mol/L}$ $[\text{NO}_2] = 5.00 \text{ mol/L}$ $[\text{NO}] = 7.20 \text{ mol/L}$ $[\text{SO}_3] = 4.00 \text{ mol/L}$

Part D: Long Answer [12 marks]

SHOW ALL YOUR WORK!!

1. A titration was run to determine the concentration of a 25.0 mL weak **polyprotic acid**, carbonic acid ($\text{H}_2\text{CO}_{3(\text{aq})}$). The titration required 55.0 mL of a 1.0 M $\text{NaOH}_{(\text{aq})}$ solution to reach the end. *For steps c & d, if you don't have a value from "a", make up a value and solve.*

Determine the concentration of the carbonic acid solution. (*hint - start by writing a balanced chemical equation*) (3 marks)

Since carbonic acid is polyprotic acid, it can give off two H^+ ions in solution. Write out both dissociation reactions for carbonic acid when it is present in solution - don't forget states and charges! (2 marks)

Using your concentration of H_2CO_3 and equations above, calculate the pH of the solution and the concentration of $[\text{HCO}_3^-]$ in the solution at equilibrium? Calculate or discuss any assumptions you have made. (4 marks)

$$K_{a1} = 4.5 \times 10^{-7}; K_{a2} = 4.7 \times 10^{-11}$$

Calculate the total percent dissociation of this weak acid. (1 mark)

How does your answer in part d prove that carbonic acid is a weak acid and not a strong acid (2 mark)